

****1. Data Cleaning:****

- Loaded the Netflix dataset and checked its shape, info, and head. - Removed duplicate records and handled missing values in key columns ('director', 'cast', 'country', 'rating'). - Converted 'date_added' to datetime format and identified missing dates. - Cleaned 'duration' column by removing text and retaining numeric values. - Saved the cleaned dataset as a new CSV file.

****2. Encoding:****

- Applied different encoding methods to transform categorical data into numeric form: 1. ****One-Hot Encoding:**** Created binary columns for 'type', 'rating', 'country', and 'genre'. 2. ****Label Encoding:**** Assigned integer values to each category. 3. ****Frequency Encoding:**** Replaced categories with their frequency ratios. 4. ****Ordinal Encoding:**** Converted categorical values into ordered numeric format.

****3. Normalization:****

- Selected all numeric columns from the encoded dataset. - Used sklearn's Normalizer to scale numerical values between 0 and 1. - This ensures equal weightage for all features during modeling.

****4. Exploratory Data Analysis (EDA):****

- Analyzed content growth over the years using line and bar plots. - Visualized the distribution of genres, ratings, and content types. - Explored the top contributing countries using bar charts and choropleth maps. - Provided insights into Netflix's global content diversity and trends.

****5. Feature Engineering:****

- Created new columns to enhance model understanding: - ****Content Origin:**** Classified shows as 'Original' or 'Licensed'. - ****Title Length:**** Measured the number of characters in titles. - ****Country Features:**** Added 'num_countries', 'is_usa', 'first_country', and top country flags. - ****Age Category:**** Grouped ratings into 'Kids', 'Family', 'Teens', and 'Adults'. - ****Length Category:**** Classified content based on duration (Short, Feature, Long, Series types). - These features improved data richness and model accuracy.

****6. Clustering (Unsupervised Learning):****

- Applied ****KMeans Clustering**** to group titles with similar characteristics. - Used features like genre, duration, and rating for clustering. - Standardized the features before clustering to ensure fair distance measurement. - Identified 5 content clusters representing similar viewing patterns.

****7. Classification (Supervised Learning):****

- Built a ****Random Forest Classifier**** to predict whether a title is a Movie or TV Show. - Used features like duration, title length, number of countries, rating, and genre. - Split the dataset into training and test sets, then standardized features. - Evaluated performance using accuracy, classification report, and confusion matrix. - Displayed feature importance to identify the strongest predictors.

****8. Insights & Visualization:****

- Visualized relationships between features using bar charts, histograms, and heatmaps. - Used Random Forest feature importance to identify key content characteristics influencing Netflix data trends. - Provided a heatmap of country vs. genre to understand regional content focus.